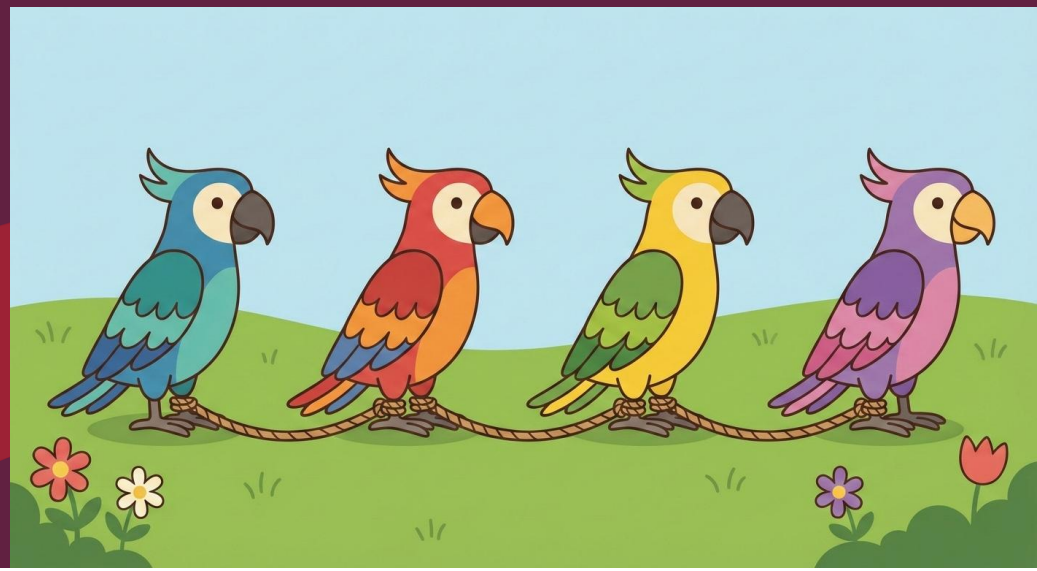
Agentic AI with LangChain and LangGraph

Presentation 2: Some Good-to-Know Foundations

Primary Instructor: Michael Z
RA: YH

Date: 2026.Jan-Feb



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0. A little bit of Review of Presentation 1

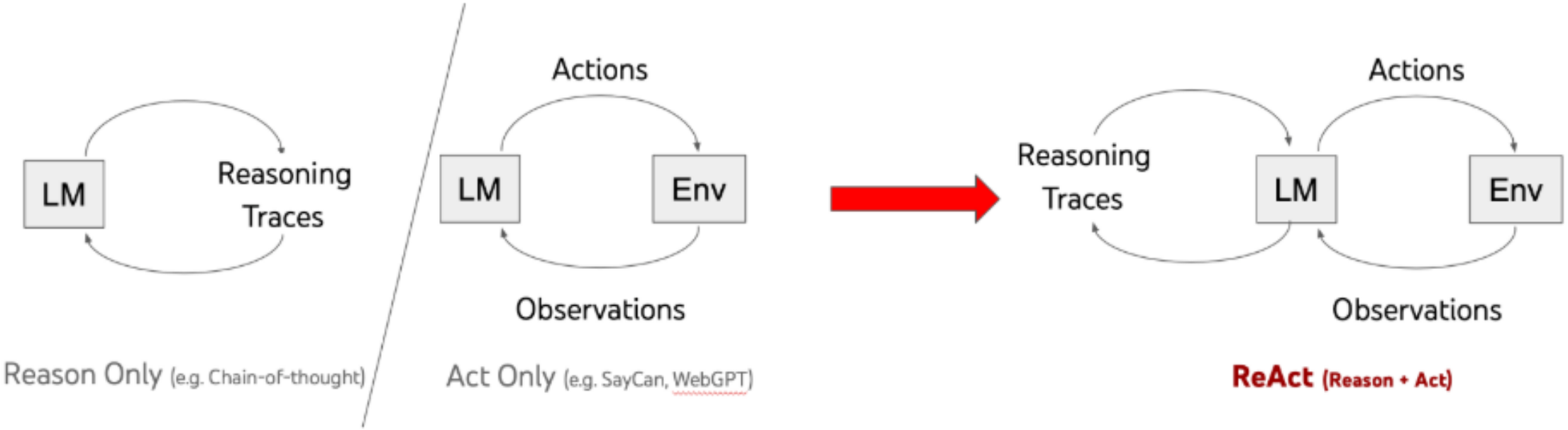
0. A little bit of Review of Presentation 1: Re and Act

I might have, or might haven't dived into the details of ReAct in presentation but here we are:

ReAct: Synergizing Reasoning and Acting in Language Models

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, Yuan Cao

[Paper] [Code] [Blogpost] [BibTex]



(DeepSeek R1?)

(I think we could watch SayCan video)

0. A little bit of Review of Presentation 1: Reasoning

LLM's reasoning capability is obtained through training (to be specific, it's post-training. Would talk about it later.). One entry from the train dataset is presented below to you:


```
[
{
"content": "Let the set $A=\\{1,2,3\\}$, and the set $B=\\{0,1,2\\}$. Then, the number of elements in $A\\cup B$ is $(\\quad)$.\n\nA: $2$\n\nB: $3$\n\nC: $4$\n\nD: $5$",
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"source": "openR1Math_extended",
"test_case": null,
"think_content": null
},
"role": "user"
},
{
"content": "<think>\nOkay, so I have this problem here about sets A and B. Let me read it again to make sure I understand. It says, \"Let the set A be {1, 2, 3}, and the set B be {0, 1, 2}. Then, the number of elements in A union B is ( ). The options are
A: 2, B: 3, C: 4, D: 5.\" Hmm, union of A and B. Right, union means all the elements that are in either set A or set B or in both. So I need to combine them and remove any duplicates. Let me write down the sets first.\n\nSet A is {1, 2, 3}. So those are the
numbers 1, 2, and 3. Set B is {0, 1, 2}. That has 0, 1, and 2. Now, if I combine these two sets, I should list all the elements from both, but make sure I don't count the same element more than once. Let me list them out.\n\nStarting with set A: 1, 2, 3.
Then set B has 0, 1, 2. So adding those to the union, we already have 1, 2, 3 from A, and from B we have 0 which isn't in A. The 1 and 2 from B are already in A, so they don't need to be added again. So combining them, the union should have elements 0, 1,
2, 3. That's four elements. Wait, but let me check again. Let me list them all out:\n\nA union B = {0, 1, 2, 3}. So the elements are 0, 1, 2, 3. Let's count them: 0 is one, 1 is two, 2 is three, 3 is four. So that's four elements. So the number of elements
is 4. The options are C: 4. So the answer should be C?\n\nWait, but let me make sure I didn't miss anything. Let me recall the formula for the number of elements in a union: |A ∪ B| = |A| + |B| - |A ∩ B|. So the size of the union is the size of A plus the
size of B minus the size of their intersection. Let's apply that.\n\nSet A has 3 elements: 1, 2, 3. Set B has 3 elements: 0, 1, 2. The intersection of A and B would be the elements common to both, which are 1 and 2. So |A ∩ B| is 2. Then |A ∪ B| = 3 + 3 - 2
= 4. Yes, that matches what I got before. So that's confirming it. Therefore, the number of elements in A union B is 4. So the answer is C.\n\nBut wait, just to double-check, maybe I made a mistake in the intersection. Let me verify the intersection. A ∩ B
is the set of elements that are in both A and B. A is {1,2,3}, B is {0,1,2}. So elements common to both are 1 and 2. So yes, two elements. So the formula gives 3 + 3 - 2 = 4. That seems correct. And listing them out gives four elements. So I think that's
right.\n\nAlternatively, maybe the answer is D:5? Wait, maybe I miscounted. Let me list the elements again. If we take all elements from A and B: A has 1,2,3; B has 0,1,2. So combining all of them: 0,1,2,3. So that's four distinct elements. There's no 4 or
anything else. So four elements. So answer is C:4. So yeah, I think that's the correct answer. The options are A:2, B:3, C:4, D:5. So the correct one is C.\n\n**Final Answer**\n\\boxed{C}\n</think>\n<answer>\n\nGiven the sets \\( A = \\{1, 2, 3\\} \\) and
\\( B = \\{0, 1, 2\\} \\), we need to find the number of elements in \\( A \\cup B \\).\n\nFirst, we recall that the union of two sets \\( A \\) and \\( B \\) includes all elements that are in \\( A \\), in \\( B \\), or in both. Listing the elements of
both sets and combining them while removing duplicates, we get:\n- Elements of \\( A \\): 1, 2, 3\n- Elements of \\( B \\): 0, 1, 2\n\nCombining these elements and removing duplicates, the union \\( A \\cup B \\) is \\( \\{0, 1, 2, 3\\} \\).\n\nTo verify,
we use the formula for the number of elements in the union of two sets:\n\\[\n|A \\cup B| = |A| + |B| - |A \\cap B|\n\\]\nwhere \\( |A| = 3 \\), \\( |B| = 3 \\), and \\( |A \\cap B| = 2 \\) (since the intersection \\( A \\cap B \\) is \\( \\{1, 2\\} \\)).
Plugging in these values:\n\\[\n|A \\cup B| = 3 + 3 - 2 = 4\n\\]\n\nThus, the number of elements in \\( A \\cup B \\) is 4, which corresponds to option C.\n\n\\[\n\\boxed{C}\n\\]</answer>",
"info": {
"answer_content": "\n\nGiven the sets \\( A = \\{1, 2, 3\\} \\) and \\( B = \\{0, 1, 2\\} \\), we need to find the number of elements in \\( A \\cup B \\).\n\nFirst, we recall that the union of two sets \\( A \\) and \\( B \\) includes all elements that are
in \\( A \\), in \\( B \\), or in both. Listing the elements of both sets and combining them while removing duplicates, we get:\n- Elements of \\( A \\): 1, 2, 3\n- Elements of \\( B \\): 0, 1, 2\n\nCombining these elements and removing duplicates, the
union \\( A \\cup B \\) is \\( \\{0, 1, 2, 3\\} \\).\n\nTo verify, we use the formula for the number of elements in the union of two sets:\n\\[\n|A \\cup B| = |A| + |B| - |A \\cap B|\n\\]\nwhere \\( |A| = 3 \\), \\( |B| = 3 \\), and \\( |A \\cap B| = 2 \\)
(since the intersection \\( A \\cap B \\) is \\( \\{1, 2\\} \\)). Plugging in these values:\n\\[\n|A \\cup B| = 3 + 3 - 2 = 4\n\\]\n\nThus, the number of elements in \\( A \\cup B \\) is 4, which corresponds to option C.\n\n\\[\n\\boxed{C}\n\\]",
"reference_answer": null,
```
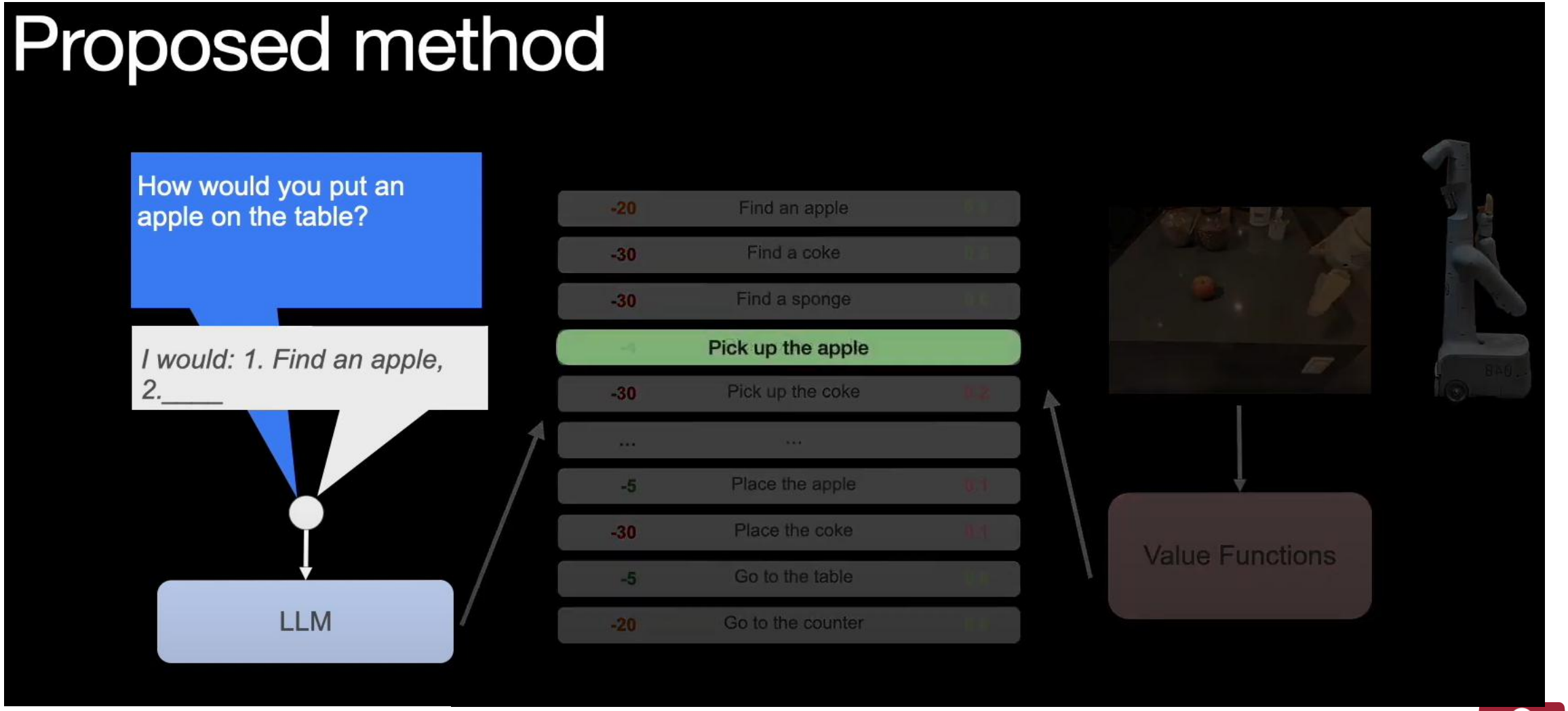

0. A little bit of Review of Presentation 1: Acting (SayCan)

Each phrase corresponds to one action. Similarly, acting ability is also obtained through training (post-training).



0. A little bit of Review of Presentation 1: ReAct

ReAct architecture requires LLMs to output tool messages in a specific format to call tools. And then, these tool messages would be parsed and sent to functions for execution. Finally, these tools would return outcomes (Observations) back to LLM.

Notably, ability of calling tools is also obtained through post-training.

Search this dataset

system
string · *lengths*
3.93k 5.08k 5%

conversations
list · *lengths*
10 12 1.8%


```
You are an expert in composing functions. You are given a question and a set of possible
functions.
Based on the question, you will need to make one or more function/tool calls to achieve the
purpose.
If none of the function can be used, point it out. If the given question lacks the parameters
required by the function,
also point it out. You should only return the function call in tools call sections.
Here is a list of functions in JSON format that you can invoke:
[{"name": "newAddress", "description": "Generates a new Ethereum address that can be used to
send or receive funds. Do not lose the password! We can't restore access to an address if you
lose it.", "parameters": {"type": "dict", "properties": {"password": {"description": "The
password for the new Ethereum address", "type": "string"}}, "required": ["password"]},
"required": null}, {"name": "Market Trends API", "description": "Get the latest market trends
and relevant news for a specified country and language.", "parameters": {"type": "dict",
"properties": {"trend_type": {"description": "Trend type.", "type": "string", "enum":
["MARKET_INDEXES", "MOST_ACTIVE", "GAINERS", "LOSERS", "CRYPTO", "CURRENCIES",
"CLIMATE_LEADERS"]}, "country": {"description": "The country for which to get trends,
specified as a 2-letter country code - see ISO 3166.", "type": "string", "default": "us"},
"language": {"description": "The language to use for the results, specified as a 2-letter
language code - see ISO 639-1.", "type": "string", "default": "en"}}, "required":
["trend_type"]}, "required": null}, {"name": "Get Futures Prices", "description": "Retrieve a
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the exchange (e.g., CME, ICE, NYMEX)", "type": "string"}, "start_string": {"description":
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and documents.", "parameters": {"type": "dict", "properties": {"identifier": {"description":
"Publicly traded company's stock symbol or Central Index Key (CIK)", "type": "string",
"default": "aapl"}}, "required": ["identifier"]}, "required": null}, {"name": "United States
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time spent away from home in the United States, providing insights into the economic impact
of the COVID-19 pandemic.", "parameters": {"type": "dict", "properties": {"string":
{"description": "The string for which to retrieve data (format: YYYY-MM-DD)", "type":
"string"}, "state": {"description": "The state for which to retrieve data (optional, default:
```

```
[
{
"from": "user",
"value": "I'm considering investing and I'd like to know what's happening in the market right
now. Could you get me the top market trends in the US?"
},
{
"from": "assistant",
"value": "[Market Trends API(trend_type=\"MARKET_INDEXES\", country=\"us\")]"
},
{
"from": "tool",
"value": "[{\"name\": \"Market Trends API\", \"results\": {\"trends\": [{\"name\": \"S&P
500\", \"description\": \"Standard & Poor's 500 Index is a market-capitalization-weighted
index of the 500 largest U.S. publicly traded companies.\", \"data\": {\"current_value\":
\"4172.80\", \"percentage_change\": \"+0.68%\"}}, {\"name\": \"DOW J\", \"description\":
\"Dow Jones Industrial Average is a price-weighted average of 30 blue-chip stocks that are
generally the leaders in their industry.\", \"data\": {\"current_value\": \"34479.60\",
\"percentage_change\": \"+0.47%\"}}, {\"name\": \"NASDAQ\", \"description\": \"The NASDAQ
Composite is a broad-based capitalization-weighted index of stocks in all three NASDAQ tiers:
Global Select, Global Market and Capital Market.\", \"data\": {\"current_value\":
\"13691.30\", \"percentage_change\": \"+0.90%\"}}]}}]"
},
{
"from": "assistant",
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& Poor's 500 Index is a market-capitalization-weighted index of the 500 largest U.S. publicly
traded companies. Its current value is 4172.80 with a percentage change of +0.68%.\n\n2.
**DOW J**: The Dow Jones Industrial Average is a price-weighted average of 30 blue-chip
stocks that are generally the leaders in their industry. Its current value is 34479.60 with a
percentage change of +0.47%.\n\n3. **NASDAQ**: The NASDAQ Composite is a broad-based
capitalization-weighted index of stocks in all three NASDAQ tiers: Global Select, Global
Market and Capital Market. Its current value is 13691.30 with a percentage change of
+0.90%.\n\nThis information can help you make informed decisions about your investment
plans."
},
{
```

0. A little bit of Review of Presentation 1: A nice chart

Input goes into LLM.

LLM might or might not reason, then would or would not take actions (call tools).

If tool is called, observation would be returned to inform reasoning or next action.

Then steps above repeat until LLM thinks it's finished. Then output is returned to user.

This is the biggest takeaway you need from this tutorial. I hope you won't forget this after you graduate!!!

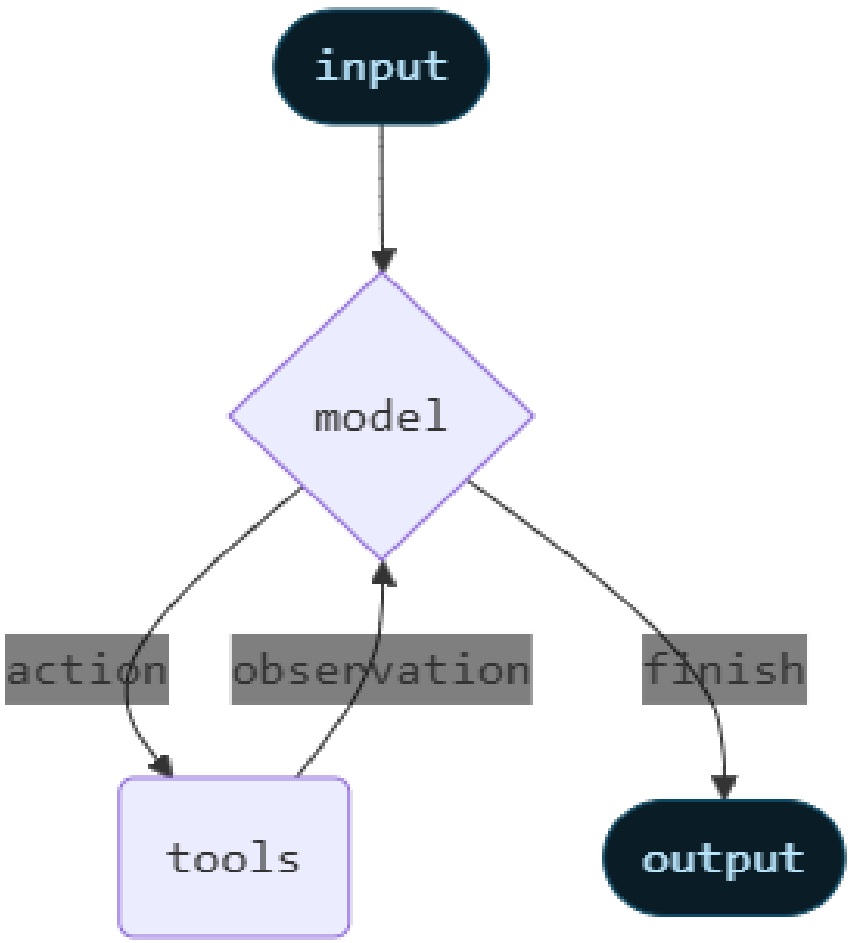


Figure 1: LangChain illustration of ReAct

Figure 1 Source: https://docs.langchain.com/oss/python/langchain/agents

(Image above was the one I copied from LangChain docs in Dec.2025 but recently it was "fixed". So, the one you could see right now is little bit different from the one I posted here. FYI.)

(Q: You blah blah blah “post-training” several times. What does it mean?
A: My bad. Here we are.)

1. How a LLM is trained from scratch?

(No worries you won’t do this in this tutorial. But feel free to do it on your own if you have GPUs!!!)

1. How a LLM is trained from scratch?: Pre-Training

Initially, a LLM is randomly initialized. It could do nothing but output nonsense, which makes sense.

Then, developers want to train it using QA pairs, which is what we think when dealing with other models. But human being don't have that many high-quality QA pairs. And hiring someone to make QA pairs is prohibitively expensive.

So, developers came up with an idea that LLM learns everything on its own first through learning how to predict next token (pretext task).

(To learn more about pretext task, please refer to self-supervised learning.)

Then, LLM could auto-complete prompts. At this moment, this is called base model.

(When you are downloading LLMs from HuggingFace, if the name of LLM doesn't contain "instruct", it's base model by default, except gpt-oss-20b and 120b. If it contains "base", it's base model.)

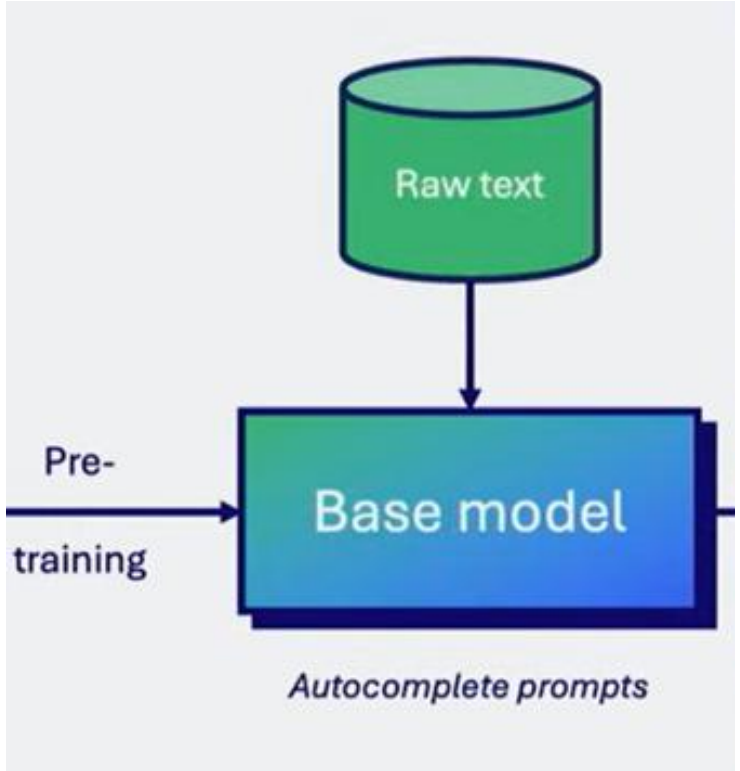


Figure 2: Pre-Training

Figure 2 Source:
https://github.com/mlabonne/llm-course

1. How a LLM is trained from scratch?: Post-Training

And then, developers train base model using QA, Instruction-Response pairs and then it would become Instruct Model.

Actually, instruct model is usable.

But to meet human preference further. Instruct model would be trained further using RL or whatever I can't name here.

(If you are prompted by ChatGPT to choose which response you like, what's taking place is alignment. But once I realized I was helping them to train model and I didn't get paid, I stopped choosing response for them 😈.)

(Since it's not a LLM course, I would not dive into details but feel free to learn LLM basics on your own.)

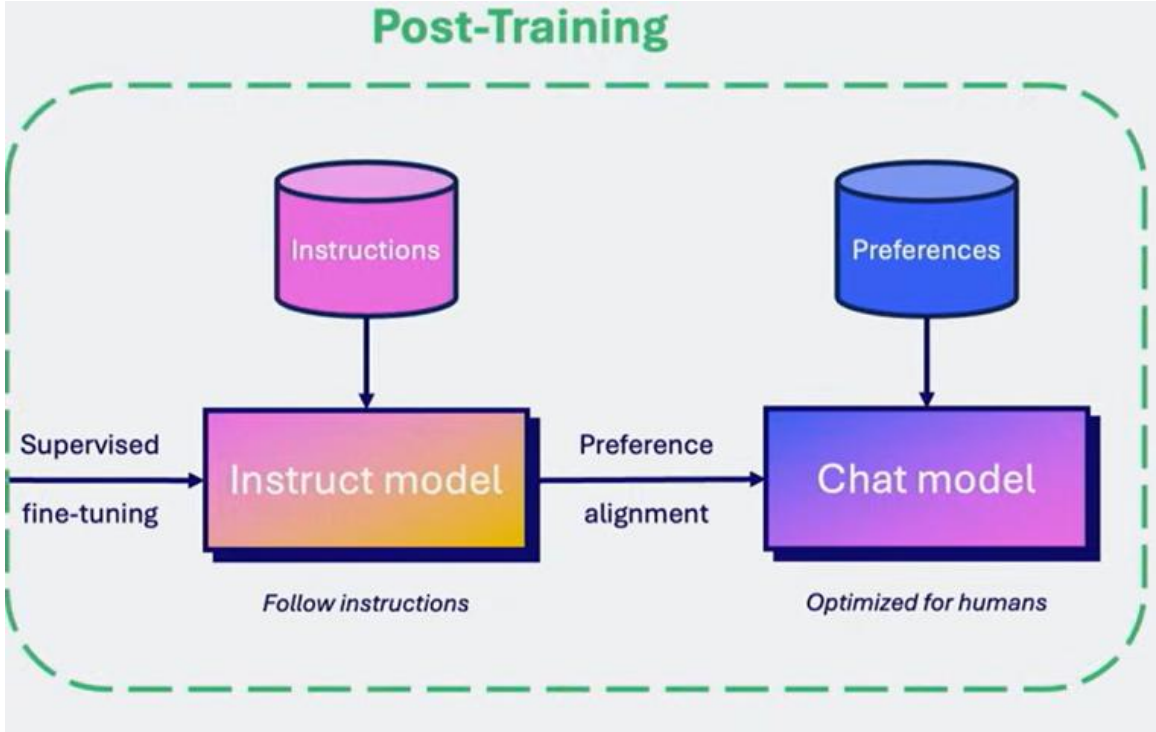


Figure 3: Post-Training

Figure 3 Source:
https://github.com/mlabonne/llm-course

1. How a LLM is trained from scratch?: An inappropriate example

(An appropriate explanation involves topics of self-supervised learning, which won't be discussed here.)

(If you have never heard about Scratch, I would open the browser and search it for you.)

2. What are the problems of tools calling after you learned how capability of tools calling is obtained?

(I think I would interact with you guys for couple mins so please speak something 👀)

(For some reasons, some vendors' LLMs don't have the capability of calling tools through API……..)

(It sounds weird, but it did once happen to me)

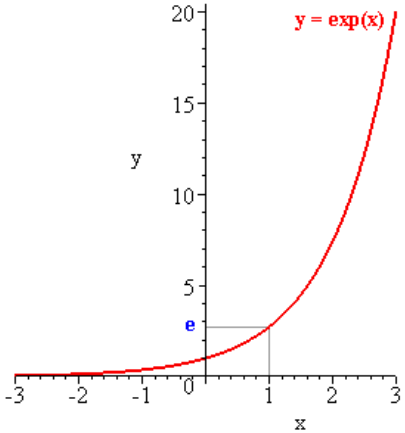
3. Some Commonly-Used Generation Config of LLM

3. Some Commonly Used Generation Config of LLM

Different models would support different parameters, but they all support these 6 parameters.

This is where temperature comes in:

$$P_T(i \mid x_{\leq t}) = \frac{e^{z_i/T}}{\sum_j e^{z_j/T}}$$



If you are interested in all parameters that could control how LLM outputs response, Please refer to Generation Configuration of HuggingFace (Don't ask me when you don't understand some of these parameters 😭).
https://huggingface.co/docs/transformers/en/main_classes/text_generation#transformers.GenerationConfig

Since we are not learning LLM, I won't dive into details a lot.

`model` `string` `required`

The name or identifier of the specific model you want to use with a provider. You can also specify both the model and its provider in a single argument using the ':' format, for example, 'openai:o1'.

`api_key` `string`

The key required for authenticating with the model's provider. This is usually issued when you sign up for access to the model. Often accessed by setting an environment variable.

`temperature` `number`

Controls the randomness of the model's output. A higher number makes responses more creative; lower ones make them more deterministic.

`max_tokens` `number`

Limits the total number of tokens in the response, effectively controlling how long the output can be.

`timeout` `number`

The maximum time (in seconds) to wait for a response from the model before canceling the request.

`max_retries` `number`

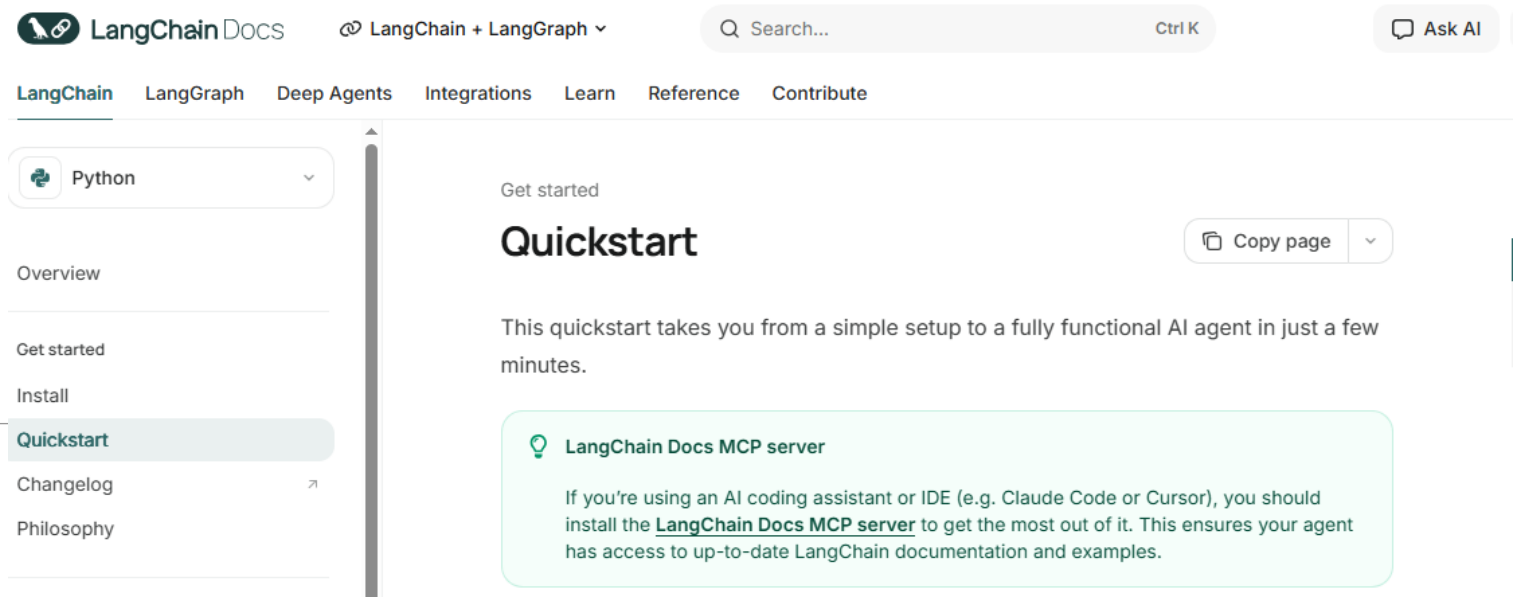
The maximum number of attempts the system will make to resend a request if it fails due to issues like network timeouts or rate limits.

4. Warm-Up

4. Warm-Up

Since we are going to dive into the fundamentals of LangChain in next presentation, you should do their QuickStart on your own.
If you find it too easy, congrats! It means you don't have to follow my pace, and you can learn it by yourself from the docs.

(Docs could be wrong and I fixed some errors for LangChain. For the most accurate information please refer to their reference.)
(Why docs != reference for LangChain? I am curious 🤔)
(If you run into any issues, just simply reach out to Yilong or me, rather than Michael.)
(We are using OpenAI API key so please simply replace Anthropic LLM with gpt-4o or others.)
(Put your API key in .env and use os.getenv() to get the value and pass it in to the init_chat_model)
(Some functions used in QuickStart would be discussed in the upcoming presentations. So, for now, just enjoy programming without actual understanding.)



4. Warm-Up: Couple of Questions btw

(If you can't answer that's okay….)

1. Is the tool decorator optional?
2. What does dataclass decorator do?
3. What are the differences between user prompt and system prompt? (open question, which means no standard answer.)
4. Can the agent call a tool if name of this tool is removed from system prompt?

…….

Thank you for your paged attention!

arXiv > cs > arXiv:2309.06180

Computer Science > Machine Learning

[Submitted on 12 Sep 2023]

Efficient Memory Management for Large Language Model Serving with PagedAttention

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, Ion Stoica